Deep Low-Rank Prior for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection is aimed at detecting observations that differ from their surroundings. To achieve this goal, low-rank models and autoencoders (AEs) have attracted a lot of attention. Although the low-rank model is self-explainable, a low-rank prior may not completely match real data. In contrast, AEs can automatically learn the discriminative features between anomalies and background, whereas AEs are not self-explainable. In this article, a deep low-rank prior-based method (DeepLR) is proposed, which combines a model-driven low-rank prior and a data-driven AE. To be specific, the low-rank prior and a fully convolutional AE architecture are incorporated through modeling an energy minimization problem solved by an iterative optimization framework, in which low-rank background estimation and network training serve as two subproblems. The low-rank background is input into the network to calculate a low-rank regularized loss, constraining the training of the network. Finally, the background can be approximately reconstructed, while the anomalies are reconstructed with significant reconstruction errors; thus, the reconstruction errors indicate the anomalous degree. The experimental results obtained on several public datasets and two large unmanned aerial vehicle (UAV)-borne datasets confirm the merit and viability of the proposed method.**

***Index Terms*— Autoencoder (AE), hyperspectral anomaly detection, iterative optimization, low-rank prior.**

NOMENCLATURE

$\mathbf{Y} \in \mathbb{R}^{(H\times W)\times B}$	Hyperspectral image.
$\mathbf{y}_{i,j} \in \mathbb{R}^{(1\times 1)\times B}$	Pixel vector at the location (i, j) of hyperspectral image $\mathbf{Y}$.
$\mathbf{Z} \in \mathbb{R}^{(H\times W)\times B}$	Uniform noise image.
$\mathbf{X} \in \mathbb{R}^{(H\times W)\times B}$	Low-rank background.
$\mathbf{x}_{i,j} \in \mathbb{R}^{(1\times 1)\times B}$	Pixel vector at the location (i, j) of low-rank background $\mathbf{X}$.
$f_\theta(\cdot)$	Network parametrized by θ.
$\mathbf{U} \in \mathbb{R}^{(H\times W)\times B}$	Scaled Lagrange multiplier.

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I. INTRODUCTION

HYPERSPECTRAL imaging allows observations of discriminative spectral information of targets of interest [1]–[3], which provides us with a powerful means for anomaly detection [4]–[6]. Hyperspectral anomaly detection has been an active area of research in hyperspectral information processing, in which anomalies refer to the observations with significant differences in spectral characteristics with their surroundings [7]–[9].

Since no *a priori* knowledge is available about anomalies [7], the background estimation is a crucial step in hyperspectral anomaly detection. The classical Reed–Xiaoli (RX) detector [10] assumes that the background obeys a Gaussian multivariate distribution, and the hyperspectral image is input to calculate a Mahalanobis distance for each pixel as the final result of the anomaly detection. Inspired by the classical RX detector, some improved methods, such as the kernel RX-algorithm [11], the linear-filter-based RX detector [12], and the blocked adaptive computationally efficient outlier nominator (BACON) detector [13], have also been proposed. The random-selection-based anomaly detector (RSAD) [14] assumes that the background obeys a more complicated chi-squared distribution, which provides a robust estimation of the background. Recently, a dual theory of likelihood ratio test (LRT)-based hyperspectral target detection has been developed for hyperspectral anomaly detection and yields various statistical LRT-based hyperspectral anomaly detectors [15]. These statistical methods are self-explainable; however, the statistical distributions may not adapt to datasets with different distributions, leading to a limited generalization capability.

In addition, the background in hyperspectral imagery usually exhibits a low-rank characteristic [16]–[23], i.e., the background can be approximately represented by several basis vectors, based on which low-rank models have been introduced into hyperspectral anomaly detection. The anomaly detection method based on low-rank and sparse representation (LRASR) [19] assumes that the background holds a low-rank property after dictionary-based sparse coding, and the anomalies are separated through matrix decomposition. In the abundance- and dictionary-based low-rank decomposition (ADLR) method [20], the background is assumed to exhibit a low-rank property in the abundance subspace after unmixing. Based on the concept that the low-rank property of the background is enhanced in local homogenous regions, the locally enhanced low-rank prior was proposed by Wang *et al.* [21], in which image segmentation is first

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performed and the low-rank property of the background is then modeled in the segmentation region. Orthogonal subspace projection (OSP) theory has also been introduced into the go-decomposition (GoDec) approach to find the low-rank and sparse matrices for anomaly-background separation [24], based on which various versions of OSP-based anomaly detectors have been developed [25], which outperformed the robust principal component analysis (RPCA) and collaborative-representation-based detector (CRD) methods. The low-rank model-based methods are self-explainable since they model the low-rank prior, which exists objectively in hyperspectral imagery, as an energy minimization problem [26]. However, the low-rank prior may not completely match real data, thus causing false alarms. Furthermore, parameters, such as the regularization parameters [27], are introduced, which are difficult to determine since no *a priori* knowledge is available for reference.

In recent years, encoder–decoder architectures, including denoising autoencoders (AEs) [28], sparse AEs [29], adversarial AEs [30], and generative adversarial networks (GANs) [31], have been applied in hyperspectral anomaly detection [32]–[40]. In the spectral constrained adversarial autoencoder (SC_AAE) [36], the hyperspectral data are preprocessed based on the consideration that the anomalies are embedded in the local smooth background, and a bilayer architecture detector is designed for anomaly detection based on the latent representation learned by the SC_AAE. The manifold-constrained AE network [35] first obtains the embedding representation of the hyperspectral data using manifold learning, which constrains the learning of the AE, and then, a detector combining the global and local reconstruction errors is used for anomaly detection using the feature maps generated by the network. The method of hyperspectral anomaly detection based on stacked denoising AEs [33] involves preprocessing the original image using principal component analysis (PCA), which requires the number of principal components to be manually set, and anomaly detection using the local Mahalanobis distance of the feature maps is then performed. The AEs can learn the discriminative features between anomalies and background, thus avoiding the assumption of a statistical distribution or image prior. However, these AE-based methods cannot realize end-to-end detection since they rely on the preprocessing [36] and detector construction procedures [35] to enhance the discriminative features between anomalies and background. More recently, an autonomous hyperspectral anomaly detection network based on fully convolutional AE has been proposed to realize end-to-end detection [41], where the background was reconstructed via the network, and an adaptive-weighted strategy was proposed to reduce the weights of reconstruction for the anomalies, whereas the adaptive-weighted strategy has its limitation. It is effective for the anomalies with very low proportions, while the performance would be reduced for the anomalies with higher proportions. Besides, the features learned from the network are not explainable.

To incorporate the advantages of a model-driven low-rank prior, which is a self-explainable handcrafted image prior, and a data-driven AE, which can automatically learn the discriminative features between anomalies and background, a deep low-rank prior-based method (DeepLR) is proposed in this article. The fully convolutional AE (FCAE) is essentially an unsupervised AE architecture, which only involves convolutional layers, in addition to the batch normalization, activation, and upsampling functions, so that it provides an efficient way to generate the reconstruction for each pixel of the image. The basic fully convolutional AE is trained based on the self-similarity criterion, and it generates the reconstruction for both the anomalies and background. To further separate the anomalies from the background in the reconstructed image, the low-rank prior of the background is introduced to constrain the training of the network. Specifically, the model-driven low-rank prior and the data-driven fully convolutional AE are jointly modeled as an energy minimization problem, which is solved through an iterative optimization framework [42]–[44]. In each iteration of the optimization, the low-rank background is first estimated via a low-rank model with nuclear regularization [45], [46]. The estimated low-rank background is then input into a fully convolutional AE with skip connections [47], [48], where a low-rank regularized loss function is proposed for the training of the network. When the iterative optimization is complete, the background can be approximately reconstructed via the proposed network, and the map of reconstruction errors can directly serve as the final anomaly detection map. As a method combining a model-driven low-rank prior and a data-driven AE, the proposed DeepLR method not only utilizes the self-learning features of an AE but also makes the detection framework self-explainable via the low-rank model.

The rest of this article is organized as follows. Section II gives a brief introduction to the related works. Section III provides a detailed description of the proposed method. Section IV describes the datasets used in this study and analyzes the experimental results. Finally, our conclusions are drawn in Section V.

II. Related Work

As mentioned above, a low-rank prior and a fully convolutional AE are incorporated through modeling an energy minimization problem, which is similar to the RPCA model. Thus, the RPCA model is first described in this section. The basic principles of AEs are then introduced.

A. Robust Principal Component Analysis

Given a data matrix $\mathbf{Y}$, it can be decomposed as [26]

$$\mathbf{Y} = \mathbf{X} + \mathbf{E} \tag{1}$$

where $\mathbf{X}$ is a low-rank matrix and $\mathbf{E}$ is a sparse matrix. $\mathbf{X}$ and $\mathbf{E}$ can then be optimized by solving the following energy minimization problem:

$$\min_{\mathbf{X},\mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_1 \quad \text{s.t.} \quad \mathbf{Y} = \mathbf{X} + \mathbf{E} \tag{2}$$

where $\|\cdot\|_*$ denotes the matrix nuclear norm, $\|\cdot\|_1$ denotes the matrix l_1-norm, and λ is the regularization parameter

In fact, the background in hyperspectral imagery is usually assumed to hold a low-rank property, i.e., the background

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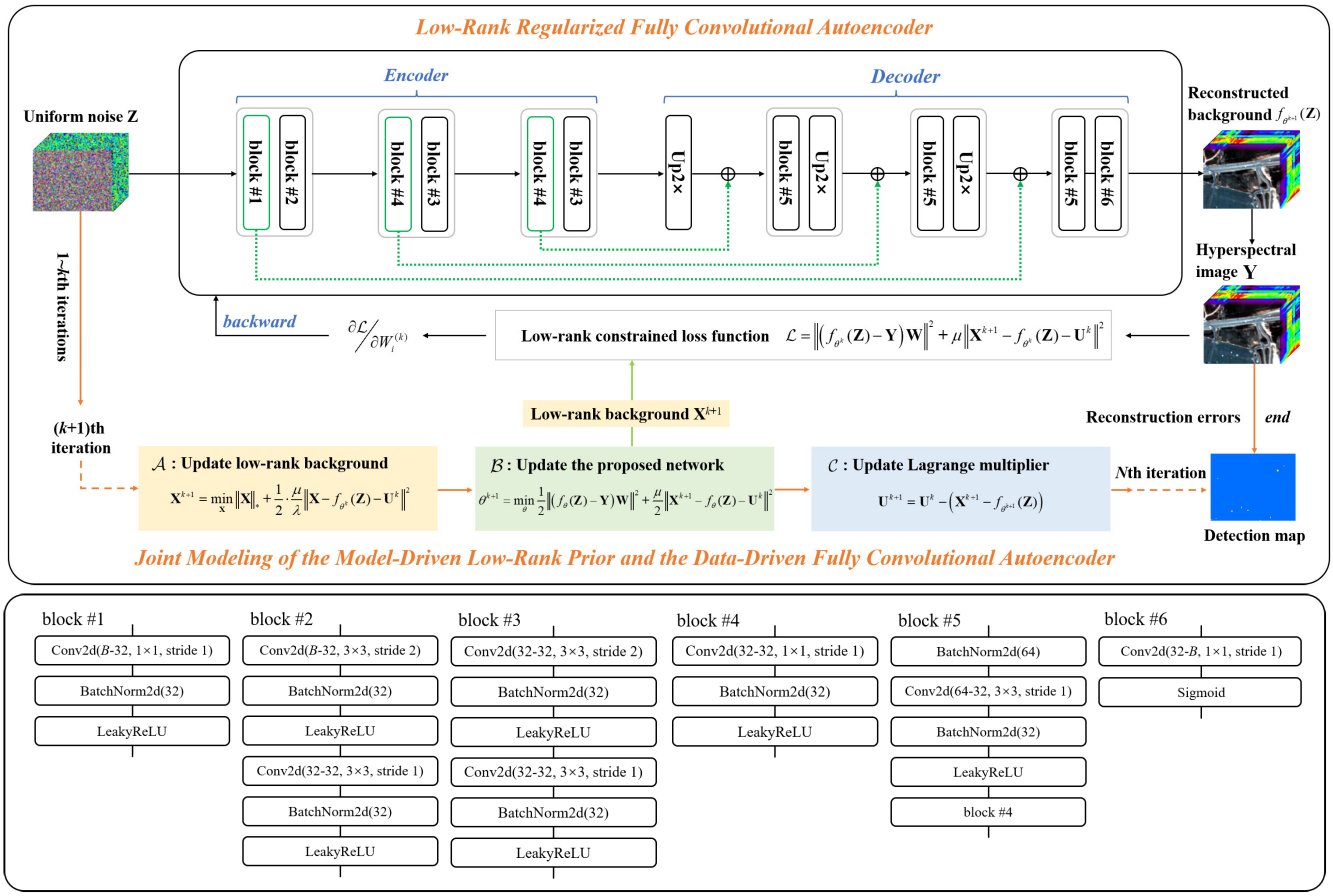


Fig. 1. Proposed low-rank regularized fully convolutional AE for hyperspectral anomaly detection.

lies in a low-dimensional subspace and can be approximately represented as a linear combination of several basis vectors. On the other hand, the anomalies exhibit sparsity, i.e., the anomalies only account for a very small proportion of the image [18]. Therefore, the hyperspectral anomaly detection problem can be solved based on the RPCA model [49].

B. Autoencoder

An AE is a feedforward artificial neural network composed of an encoder and a decoder. The encoder is used to learn the mapping from the input layer $\mathbf{x}$ to the hidden layer $\mathbf{y}$, and the decoder is used to learn the mapping from the hidden layer $\mathbf{y}$ to the output layer $\tilde{\mathbf{x}}$, which can be mathematically depicted as follows:

$$\mathbf{y} = f(\mathbf{x}; \mathbf{W}, \mathbf{b}) \tag{3}$$

$$\tilde{\mathbf{x}} = f\left(\mathbf{y}; \tilde{\mathbf{W}}, \tilde{\mathbf{b}}\right) \tag{4}$$

where the weights $\mathbf{W}$ and $\tilde{\mathbf{W}}$ and biases $\mathbf{b}$ and $\tilde{\mathbf{b}}$ can be estimated by a backpropagation method [50]–[52] and f represents the activation function. The AE is aimed at generating an output approximating the input based on the self-similarity criterion. Thus, the basic loss function is the root-mean-square error

$$\mathcal{L}_1 = \|\mathbf{x} - \tilde{\mathbf{x}}\|_2. \tag{5}$$

III. Deep Low-Rank Prior for Hyperspectral Anomaly Detection

In this article, to incorporate the advantages of a model-driven low-rank prior and a data-driven AE, a DeepLR is proposed. To be specific, the low-rank prior and a fully convolutional AE are incorporated by modeling an energy minimization problem, which is solved through the alternating direction method of multiplier (ADMM) iterative optimization framework. When the iterative optimization is complete, the background can be approximately reconstructed, while the anomalies are reconstructed with significant reconstruction errors; thus, the reconstruction errors indicate the anomalous degree.

The flowchart of the proposed DeepLR method is shown in Fig. 1. Accordingly, the proposed DeepLR method can be described in three main parts: 1) the overall anomaly detection framework via iteration optimization; 2) the low-rank model with nuclear regularization for background estimation; and 3) the low-rank regularized fully convolutional AE for end-to-end detection. To allow a better understanding of the proposed method, a summary of the notation used in this article is provided in the Nomenclature.

A. Joint Modeling of the Model-Driven Low-Rank Prior and the Data-Driven Fully Convolutional AE

It has been demonstrated that a great deal of image statistics can be captured in the parameter space of a deep neural

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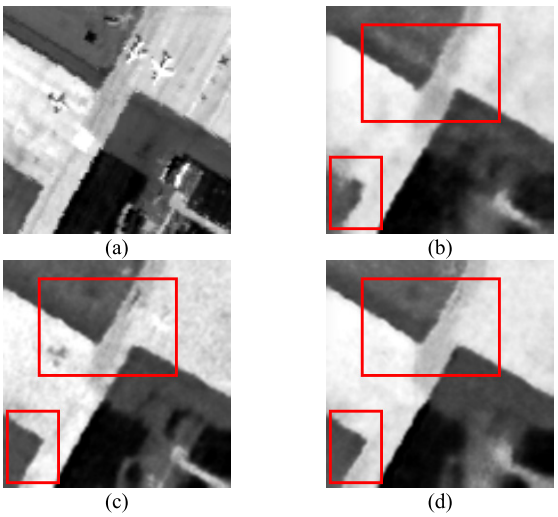


Fig. 2. Illustration of the introduction of the low-rank prior and adaptive-weighting strategy. (a) Original image. (b) Without the low-rank prior and weighting. (c) With the low-rank prior but without the weighting. (d) With the low-rank prior and weighting.

network [47], [48]. It is possible to approximately reconstruct the background of hyperspectral imagery via an AE structure. Accordingly, the reconstruction errors can be utilized to indicate the anomalous degree, thus realizing end-to-end detection. Therefore, the proposed method uses the network in [48] as the baseline network, while the layers of network and the dimensionality of feature map have been decreased, to reduce the computation cost.

However, the basic network tends to reconstruct both the anomalies and background since it is based on the self-similarity criterion, in which case the features of the anomalies will be weakened. To this end, Auto-AD [41] proposed an adaptive-weighting strategy, which assumed that the anomalies are difficult to be reconstructed by the network due to the small-size and low-probability property. In this case, the anomalies appear large reconstruction errors in the early epochs of network training. The large reconstruction errors can be transformed into small weights to reduce the weights of reconstruction for the anomalies.

The adaptive-weighted strategy has its limitation. It is effective for the anomalies with very low proportions, while the performance would be reduced for the anomalies with higher proportions. To solve this problem, the proposed method introduces the low-rank prior. Although the background estimated by the low-rank regularized model is not pure, the responses of anomalies have been weakened compared with the original image. In this way, the network is prevented from reconstructing the anomalies to some extent, especially for the anomalies with higher proportions.

As shown in Fig. 2(b) and (d), the reconstructed background with the low-rank prior exhibits more spatial details than that without the low-rank prior. Comparing Fig. 2(c) and (d), it can be seen that more pure background is reconstructed with the introduction of the adaptive-weighting strategy.

In this article, considering the following two cases, the low-rank prior and a fully convolutional AE are incorporated by modeling an energy minimization problem, which can be mathematically depicted as follows:

$$\begin{aligned} &\min_{\mathbf{X},\theta} \frac{1}{2}\|(f_\theta(\mathbf{Z})-\mathbf{Y})\mathbf{W}\|^2+\lambda\|\mathbf{X}\|_* \\ &\text{s.t. } \mathbf{X}=f_\theta(\mathbf{Z}) \end{aligned} \tag{6}$$

where $\mathbf{Z}$ has the same dimensionality as the hyperspectral image. As in the generative model of the GAN, the network $f_\theta(\cdot)$ uses the uniform distribution as a prior to generate a reconstruction of the background. λ denotes the regularization parameter. $\mathbf{W}$ is the adaptive-weighted map, which is described in the following.

The energy minimization problem in (6) can be solved via the ADMM [42]. The constraint in (6) can be turned into a penalty using the augmented Lagrangian [53], which is defined as follows:

$$\begin{aligned} \min_{\mathbf{X},\theta} \frac{1}{2}\|(f_\theta(\mathbf{Z})-\mathbf{Y})\mathbf{W}\|^2+\lambda\|\mathbf{X}\|_*+\frac{\mu}{2}\|\mathbf{X}-f_\theta(\mathbf{Z})\|^2 \\ +\mathbf{V}^T(\mathbf{X}-f_\theta(\mathbf{Z})) \end{aligned} \tag{7}$$

where $\mathbf{V}$ stands for the Lagrange multiplier and μ is a penalty parameter. The scaled form is given as follows:

$$\begin{aligned} \min_{\mathbf{X},\theta} \frac{1}{2}\|(f_\theta(\mathbf{Z})-\mathbf{Y})\mathbf{W}\|^2+\lambda\|\mathbf{X}\|_* \\ +\frac{\mu}{2}\|\mathbf{X}-f_\theta(\mathbf{Z})-\mathbf{U}\|^2-\frac{\mu}{2}\|\mathbf{U}\|^2. \end{aligned} \tag{8}$$

The optimization problem in (8) can be split into three subproblems, which are solved alternately in an iterative procedure. In the kth iteration

$$\begin{cases} \mathbf{X}^{k+1}=\min_{\mathbf{X}}\|\mathbf{X}\|_*+\dfrac{1}{2}\cdot\dfrac{\mu}{\lambda}\|\mathbf{X}-f_{\theta^k}(\mathbf{Z})-\mathbf{U}^k\|^2, & \mathcal{A} \\ \theta^{k+1}=\min_\theta \dfrac{1}{2}\|(f_\theta(\mathbf{Z})-\mathbf{Y})\mathbf{W}\|^2 \\ \qquad +\dfrac{\mu}{2}\left\|\mathbf{X}^{k+1}-f_\theta(\mathbf{Z})-\mathbf{U}^k\right\|^2, & \mathcal{B} \\ \mathbf{U}^{k+1}=\mathbf{U}^k-\left(\mathbf{X}^{k+1}-f_{\theta^{k+1}}(\mathbf{Z})\right), & \mathcal{C}. \end{cases} \tag{9}$$

1) *To Solve Subproblem* $\mathcal{A}$: Fix $f_\theta(\mathbf{Z})$ and $\mathbf{U}$, and the low-rank background $\mathbf{X}$ can be updated using (10)–(12)

$$\mathbf{X}^{k+1}=\mathcal{D}_{\lambda/\mu}\left(f_{\theta^k}(\mathbf{Z})+\mathbf{U}^k\right) \tag{10}$$

where $\mathcal{D}_{\lambda/\mu}(\cdot)$ is the singular value thresholding operator [54], which can be mathematically depicted as follows:

$$\mathcal{D}_{\lambda/\mu}(\mathbf{G})=\mathbf{P}\mathcal{S}_{\lambda/\mu}\left(\sum\right)\mathbf{Q}^T \tag{11}$$

where $\mathbf{P}\sum\mathbf{Q}^T$ is the singular value decomposition of the matrix $\mathbf{G}=f_{\theta^k}(\mathbf{Z})+\mathbf{U}^k$ and $\mathcal{S}_{\lambda/\mu}(\cdot)$ is the shrinkage operator, which is defined as follows:

$$\mathcal{S}_{\lambda/\mu}(x)=\operatorname{sgn}(x)\max(|x|-\lambda/\mu,0) \tag{12}$$

$\mathcal{S}_{\lambda/\mu}(\sum)$ is used to implement the shrinkage operator on each diagonal element of $\sum$.

2) *To Solve Subproblem* $\mathcal{B}$: Fix $\mathbf{U}$ and $\mathbf{X}$ and update the network $f_\theta(\mathbf{Z})$. The objective function is then

$$\begin{aligned} \theta^{k+1}=\min_\theta \frac{1}{2}\|(f_\theta(\mathbf{Z})-\mathbf{Y})\mathbf{W}\|^2 \\ +\frac{\mu}{2}\left\|\mathbf{X}^{k+1}-f_\theta(\mathbf{Z})-\mathbf{U}^k\right\|^2. \end{aligned} \tag{13}$$

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An update of $f_\theta(\mathbf{Z})$ is equivalent to training the AE one time. As shown in (13), when training the network for background reconstruction, the low-rank prior of the background is considered with the introduction of $\mathbf{X}^{k+1}$. The solution of (13) and the structure of the proposed network are further discussed in the next part of this section.

3) *To Solve Subproblem* $\mathcal{C}$: Fix $\mathbf{X}$ and $f_\theta(\mathbf{Z})$ and the Lagrange multiplier $\mathbf{U}$ can be updated using (14)

$$\mathbf{U}^{k+1} = \mathbf{U} - \left(\mathbf{X}^{k+1} - f_{\theta^{k+1}}(\mathbf{Z})\right). \tag{14}$$

When the optimization is completed, the background will be reconstructed by the proposed network

$$\mathbf{X}^N = f_{\theta^N}(\mathbf{Z}) \tag{15}$$

and the reconstruction errors can be utilized to indicate the anomalous degree

$$D\left(\mathbf{y}_{i,j}\right) = \left\|\mathbf{y}_{i,j} - \mathbf{x}_{i,j}^N\right\|^2 \tag{16}$$

where $\mathbf{x}_{i,j}^N$ denotes the pixel vector at location (i, j) of reconstructed background $\mathbf{X}^N$.

Compared with simply training the network, the additional steps of the ADMM include the solutions to subproblems A and X, where the increase of the computational complexity mainly comes from the solution to subproblem A, i.e., the calculation of (10), but the overall increase of the computational complexity is not significant.

B. Low-Rank Regularized Fully Convolutional AE

Specifically, a fully convolutional AE with skip connections and a low-rank regularized loss function are proposed to reconstruct the background. The parameters of the network θ are updated by solving the optimization problem in (13), which is equivalent to training the network with the following loss function, which is referred to as the low-rank regularized loss function in this article:

$$\mathcal{L} = \|(f_{\theta^k}(\mathbf{Z}) - \mathbf{Y})\mathbf{W}\|^2 + \mu\left\|\mathbf{X}^{k+1} - f_{\theta^k}(\mathbf{Z}) - \mathbf{U}^k\right\|^2. \tag{17}$$

The term $\|(f_{\theta^k}(\mathbf{Z}) - \mathbf{Y})\mathbf{W}\|^2$ is actually the traditional loss function of the AE based on the principle of self-similarity. The term $\mu\|\mathbf{X}^{k+1} - f_{\theta^k}(\mathbf{Z}) - \mathbf{U}^k\|^2$ can be seen as a penalty term, which constrains the network to approximate the low-rank background.

The network architecture is composed of an encoder and a decoder, as shown in Fig. 1. To simplify the flowchart, seven blocks are used to represent the network architecture, where each block contains a convolutional layer.

1) Encoder: The encoder contains nine convolutional layers, with every convolutional layer followed by batch normalization [55] and a leaky rectified linear unit (LeakyReLU) activation function. Blocks #1 and #4 contain a 1×1 convolutional layer with a stride of 1. The feature maps generated by blocks #1 and #4 are not input into the next convolutional layer but are concatenated with the feature maps of the corresponding layers of the decoder through skip connections, as denoted by the green lines in Fig. 1. During decoding, the skip connections complement the features with spatial details in the earlier layers of the network [56], thus improving the spatial accuracy of the reconstructed background. In the encoder, except for the convolutional layer in block #1 and the first convolutional layer in block #2, which decrease the dimensionality of the hyperspectral image and generate a feature map with 32 dimensions, the other convolutional layers do not decrease the dimensionality of the feature maps. This means that the dimensionality of the feature maps remains unchanged at 32 during the encoding process. In this way, the spectral features are preserved. Note that the network aims to reconstruct the background so that the features of the anomalies should be suppressed. The anomalies in hyperspectral imagery are usually small in size, and the number of anomalous pixels is very limited. Thus, the convolutional layers in blocks #2 and #3 perform spatial downsampling using a 3×3 convolutional layer with a stride of 2. The features of the anomalies are weakened in the feature maps during the encoding process. In each blocks #2 and block #3, a 3×3 convolutional layer with a stride of 1 follows the first convolutional layer.

2) Decoder: The decoder contains seven convolutional layers. Unlike the encoder, the decoder performs upsampling using nearest neighbor interpolation with the scale of 2, as shown in Fig. 1. The input of every block #5 is a feature map with 64 dimensions, i.e., two feature maps with 32 dimensions are concatenated via the skip connections. Thus, in each block #5, a 3×3 convolutional layer follows the batch normalization, and the input feature map with 64 dimensions is decreased to 32 dimensions. Block #4 then follows the other batch normalization and the LeakyReLU activation in each block #5. The last block #6 contains a 1×1 convolutional layer with a stride of 1, through which the feature map is increased from 32 dimensions to the same dimensionality as the original hyperspectral image. Unlike the other blocks, the convolutional layer in block #6 is followed by a sigmoid activation function.

Another issue is how to generate the adaptive-weighted map, which is based on the concept that the anomalies exhibit larger reconstruction errors than the background. The reconstruction error of each pixel can be calculated as follows:

$$e_{i,j}^k = \left\|\mathbf{y}_{i,j} - \tilde{\mathbf{y}}_{i,j}^k\right\|_2 \tag{18}$$

where $\tilde{\mathbf{y}}_{i,j}^k$ denotes the reconstructed pixel vector. We then obtain the map of reconstruction errors $\mathbf{E}$ combining the reconstruction errors of all the pixels.

The reconstruction error of each pixel can be transformed into a weight using (19), and we obtain the adaptive-weighted map $\mathbf{W}$ combining all the weights of all the pixels

$$w_{i,j} = \max(\mathbf{E}) - e_{i,j}^k. \tag{19}$$

In the adaptive-weighted map $\mathbf{W}$, the anomalies have smaller weights than the background, thus preventing the anomalies from being reconstructed. Furthermore, the weights are adaptive since they are derived from the reconstruction errors and are adaptively updated using (18)–(19) every 50 iterations.

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Algorithm 1 DeepLR

Input: hyperspectral image $\mathbf{Y} \in \mathbb{R}^{(H\times W)\times B}$

Initialization: uniform noise image $\mathbf{Z} \in \mathbb{R}^{(H\times W)\times B}$, $\lambda = 1.0$, $\mu = 0.5$, $\mathbf{U} = \mathbf{0} \in \mathbb{R}^{(H\times W)\times B}$, $\mathbf{W} = \mathbf{0} \in \mathbb{R}^{H\times W}$

ADMM iterative optimization:

1: Update the low-rank background $\mathbf{X}^{k+1}$ by solving (10);
2: Update the adaptive-weighted map $\mathbf{W}$ using (18)–(19) every 50 iterations;
3: Update the network $f_{\theta^{k+1}}$ by solving (13);
4: Update the Lagrange multiplier $\mathbf{U}$ by solving (14);
5: Check the convergence condition: $\frac{1}{50}\sum_{i=k}^{k+50}(\mathcal{L}^{i+1} - \mathcal{L}^{i}) < \delta$

Generate anomaly detection map using (16)

Output: Anomaly detection map

C. Implementation Issues

The hyperspectral image should be normalized in the range of (0, 1] before being input into the network. The normalization method is very simple, i.e., the values of each pixel vector are divided by the maximum value of the image. The input of the network is an image filled with uniform noise, which has the same dimensionality as the input hyperspectral image, i.e., $\mathbf{Z} \in \mathbb{R}^{(H\times W)\times B}$. The values of the input image with uniform noise are all sampled from a uniform distribution in the range [0, 0.1]. The adaptive-weighted map $\mathbf{W}$ is initialized as a zero tensor, i.e., $\mathbf{W} = \mathbf{0} \in \mathbb{R}^{H\times W}$.

The ADAM algorithm [50] is adopted for training. The final output of the network is an image of a size equal to the size of the input hyperspectral image, which is referred to as the reconstructed background image.

The training of the proposed network stops when the average variation of the loss is below δ within the last 50 iterations, where k denotes the number of iterations. The threshold δ can generally be set as either 0.0001 or 0.00001

$$\frac{1}{50}\sum_{i=k}^{k+50}\left(\mathcal{L}^{i+1} - \mathcal{L}^{i}\right) < \delta. \tag{20}$$

The Lagrange multiplier is initialized as a zero matrix with the same dimensionality as the hyperspectral image. The regularization parameter λ was set to 1.0 in the experiments conducted in this study, and the penalty parameter μ was set to 0.5 for all the experimental datasets. It is worth mentioning that the penalty parameter can be set as $\mu = \rho\mu$, which will be gradually increased in the iterations to speed up the convergence, where ρ can be set slightly larger than 1, such as 1.1.

The proposed DeepLR method achieves end-to-end anomaly detection. The input is the hyperspectral image, without any preprocessing procedures. The map of reconstruction errors of the network is directly utilized for anomaly detection, without any postprocessing procedures such as detector construction.

The proposed DeepLR method is described in detail in Algorithm 1.

IV. Experiments and Analyses

The proposed DeepLR method was tested on two public airborne hyperspectral datasets and two large unmanned aerial vehicle (UAV)-borne hyperspectral datasets (600 × 1800 pixels and 3600 × 600 pixels, respectively). The comparative algorithms were the global RX detector (GRX) [10], the anomaly detection method based on LRASR [19], the ADLR method [20], the CRD method [57], the SC_AAE [36], and the autonomous hyperspectral anomaly detection network based on fully convolutional AE (Auto-AD) [41]. To verify the effectiveness of the proposed low-rank prior, the results of an adaptive-weighted fully convolutional AE (W-FCAE) [48] are also compared with those of DeepLR. W-FCAE has the same network architecture as DeepLR, just without the introduction of the low-rank prior.

A. Experimental Settings

1) Dataset Description: The first dataset was collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) [58] over a suburban residential area of MI, USA, and is referred to as the airborne HYDICE dataset in this article. This dataset contained 162 spectral bands after removing the bands of the water absorption regions, low signal-to-noise ratio, and poor quality, covering the spectral range of 400–2500 nm. The image scene was of 80 × 100 pixels, and the spatial resolution was 3 m/ pixel. The spectra of the anomalies and background materials are given in Fig. 3(a). Ten man-made vehicles were considered as anomalies, with a total number of 17 pixels, and the background land-cover types were vegetation, soil, water, and roads.

The second dataset, which we refer to as AVIRIS-1 [21], was a subset of an image collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. This dataset covered the spectral range of 400–2500 nm, with 186 spectral bands, and the spatial resolution was 3.5 m/pixel. The image scene was of 100 × 100 pixels in size. Three planes were considered as anomalies, which accounted for 1.43% of the image. The spectra of the anomalies and background materials are given in Fig. 3(b).

The third dataset—the Wuhan University UAV-borne hyperspectral image (WHU-Hi) park dataset (WHU-Hi-Park dataset)—was collected by the Intelligent Data Extraction, Analysis and Application of Remote Sensing (RSIDEA) Group, Wuhan University, Wuhan, China. This dataset was obtained by a Nano-Hyperspec visible and near-infrared hyperspectral sensor mounted on a DJI M600 six-rotor UAV platform [59], [60]. The main technical specifications of this dataset are listed in Table I. The sensor provides 270 spectral bands covering the spectral range of 400–1000 nm. The data collection was carried out on March 27, 2019, over a city park in Zhengzhou, Henan, China. The flight height of the UAV system was 100 m so that the spatial resolution reaches 8 cm/pixel. This dataset is much larger than the most commonly used hyperspectral anomaly detection datasets, with 600 × 1800 pixels in total. The spectra of the anomalies and background materials are given in Fig. 3(c). The anomalies are listed in Table II, including pedestrian, plastic planes, and

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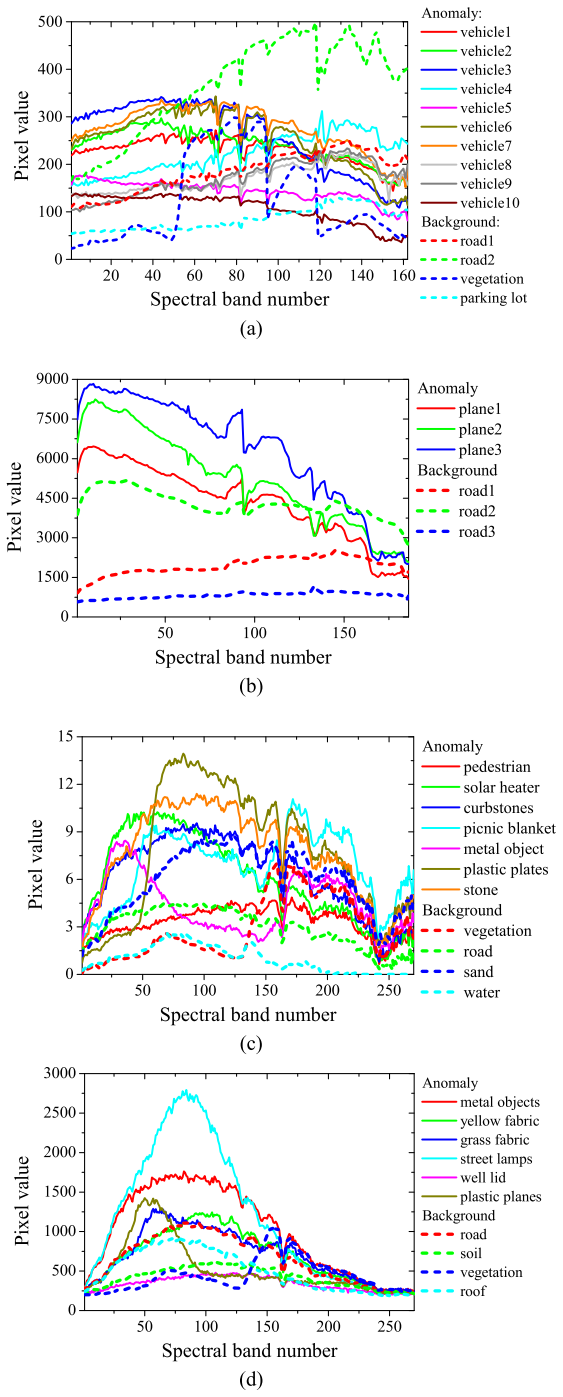


Fig. 3. Spectra of the anomalies and background materials. (a) HYDICE dataset. (b) AVIRIS-1 dataset. (c) WHU-Hi-Park dataset. (d) WHU-Hi-Station dataset.

curbstones, with a total number of 1388 pixels, as shown in Fig. 6(b). The WHU-Hi-Park dataset was split into 200 × 200 blocks for the experiments. The variation in detection performance between the different blocks can therefore reflect the generalization capability of the methods.

The fourth dataset—the WHU-Hi-Station dataset—was also collected by the RSIDEA Group using the same UAV-borne hyperspectral imaging system. The data collection was carried out on January 17, 2019, over a satellite Earth station in

TABLE I

MAIN SPECIFICATIONS OF THE DATA COLLECTION CAMPAIGN FOR THE WHU-HI-PARK AND WHU-HI-STATION DATASET

Wavelength	400–1000 nm
Spectral bands	270
Spectral resolution	2.2 nm
Digitization	12 bit
Focal length	8 mm
Flight height	100 m
Spatial resolution	8 cm

TABLE II

DESCRIPTION OF THE ANOMALIES IN THE WHU-HI-PARK DATASET

Anomaly	Description	Pixels	Ratio
T1	Trash can	40	0.0037%
P1, P2	Plastic planes	244	0.0226%
S1	Stone	21	0.0019%
M1	Metal object	176	0.0163%
B1	Picnic blanket	552	0.0511%
K1, K2, K3, K4	Curbstones	242	0.0224%
H1	Solar heater	40	0.0037%
PP	Pedestrian	73	0.0068%

TABLE III

DESCRIPTION OF THE ANOMALIES IN THE WHU-HI STATION DATASET

Anomaly		Description	Pixels	Ratio
M1, M2, M3, M4, M5, M6, M7		Metal objects	352	0.0163%
C	C1, C2	Objects camouflaged with yellow fabric	199	0.0092%
	C3, C4	Objects camouflaged with green fabric		
L1, L2		Streetlamps	145	0.0067%
W1		Well lid	112	0.0052%
P1, P2, P3		Plastic plates	264	0.0122%

Wuhan, Hubei, China. The flight height was 50 m so that the spatial resolution of the acquired image is 4 cm/pixel. This dataset is even larger than the WHU-Hi-Park dataset, with 3600 × 600 pixels. The spectra of the anomalies and background materials are given in Fig. 3(d). The anomalies are listed in Table III and include metal objects, objects camouflaged with yellow fabric or green fabric, and street lamps, with a total number of 1072 pixels, as shown in Fig. 7(b). This dataset was also split into 200 × 200 blocks to test the generalization capability of the methods.

2) Parameter Settings: The parameters of LRASR were set according to the description in the original article, i.e., the number of clusters K and the number of selected pixels P were set to 15 and 20, respectively. The regularization parameters β and λ were set to 0.1 for all four experimental datasets. CRD involves three parameters: the inner window size w_{in}, the outer window size w_{out}, and the regularization parameter λ. According to the experiments described in the original article, CRD achieved the highest detection accuracy on all the experimental datasets when λ was set to 10^{-6}. Therefore, λ was set to 10^{-6} in this study. (w_{in}, w_{out}) were set as (7, 15), (11, 17), (37, 41), and (35, 39) for the HYDICE,

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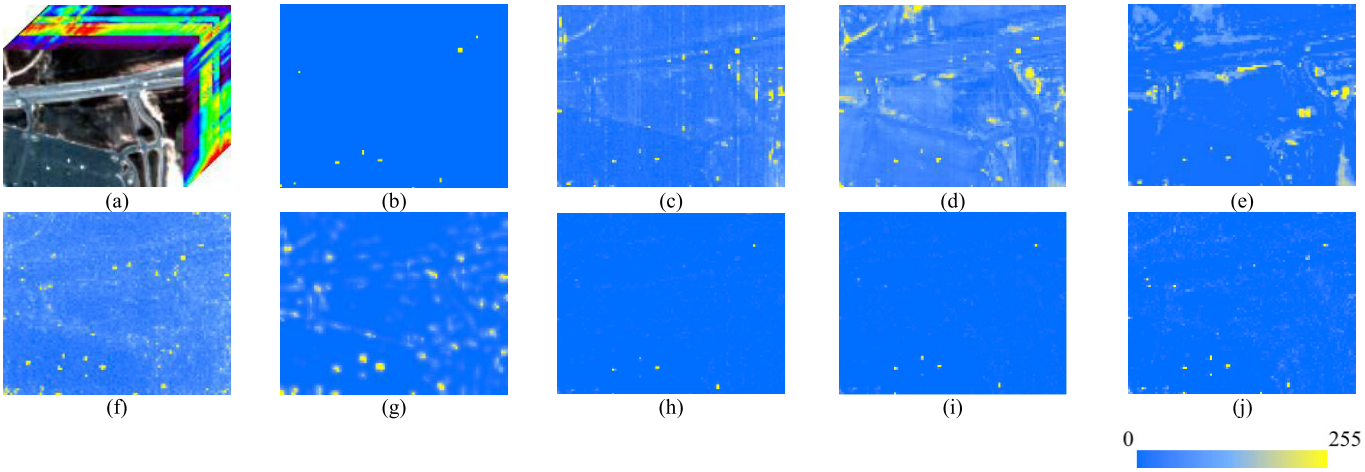


Fig. 4. Airborne HYDICE dataset and the anomaly detection results. (a) Data cube. (b) Ground-truth map. (c) GRX. (d) LRASR. (e) ADLR. (f) CRD. (g) SC_AAE. (h) Auto-AD. (i) W-FCAE. (j) DeepLR.

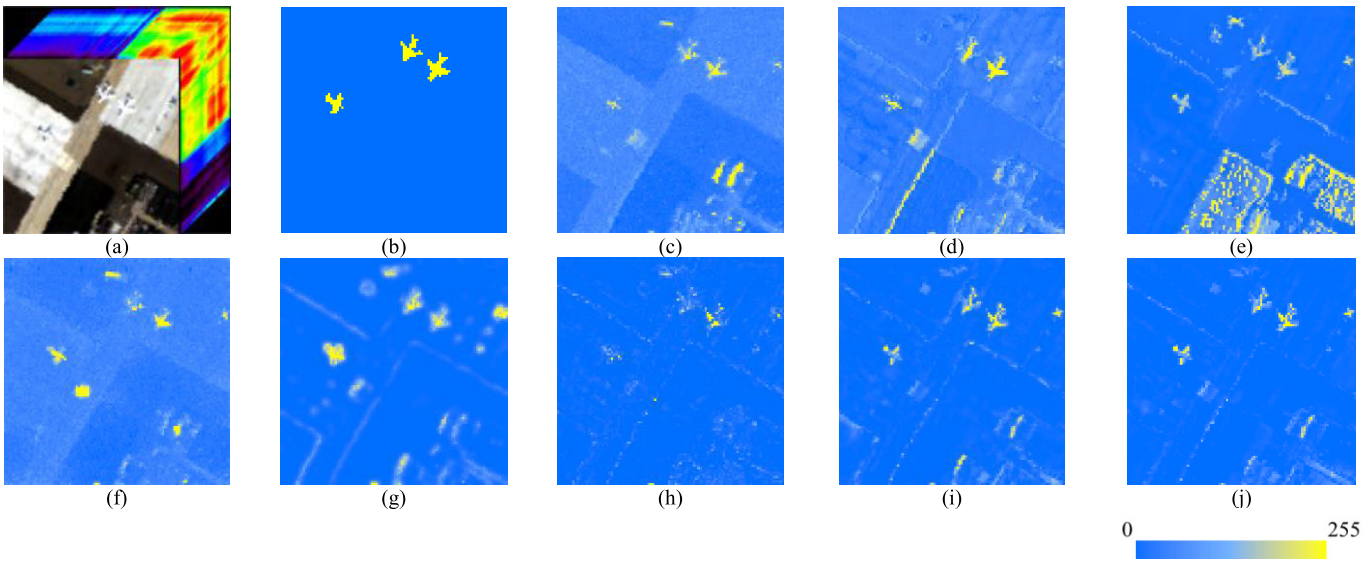


Fig. 5. Airborne AVIRIS-1 dataset and the anomaly detection results. (a) Data cube. (b) Ground-truth map. (c) GRX. (d) LRASR. (e) ADLR. (f) CRD. (g) SC_AAE. (h) Auto-AD. (i) W-FCAE. (j) DeepLR.

AVIRIS-1, WHU-Hi-Park, and WHU-Hi-Station datasets, respectively, after the parameter adjustment experiments. The threshold δ of W-FCAE and DeepLR was set as 0.00001 for the HYDICE dataset and 0.0001 for the other datasets.

B. Experimental Results

In this section, the detection results of the different methods are qualitatively analyzed through 2-D plots of the obtained detection results, which are all linearly stretched to [0, 255] for display.

1) Airborne HYDICE Dataset: The detection results for the HYDICE dataset are shown in Fig. 4, where the result of the proposed DeepLR method exhibits fewer background false alarms than those of GRX, LRASR, ADLR, and CRD methods. The backgrounds in the results of LRASR and ADLR show stronger responses than in the results of the other methods. There exist slightly more false alarms in the result of SC_AAE than that of DeepLR, while the anomalies exhibit stronger responses. Comparing the results of Auto-AD, W-FCAE, and DeepLR, although the results of Auto-AD and W-FCAE exhibit better background suppression performance, some anomalies are with low responses.

2) Airborne AVIRIS-1 Dataset: The detection results for the AVIRIS-1 dataset are shown in Fig. 5(c)–(g), where the responses of the anomalies are stronger for the proposed DeepLR method than for GRX, Auto-AD, and CRD. Specifically, the anomalies in the results of GRX, Auto-AD, and CRD are not complete visually, while the anomalies appear more intact in the result DeepLR. There are obvious false alarms in the results of LRASR and ADLR. Comparing the results of SC_AAE and DeepLR, in the result of SC_AAE, the anomalies exhibit stronger responses, while there exist more false alarms. Compared with the result of W-FCAE, the result of the proposed DeepLR method exhibits a better

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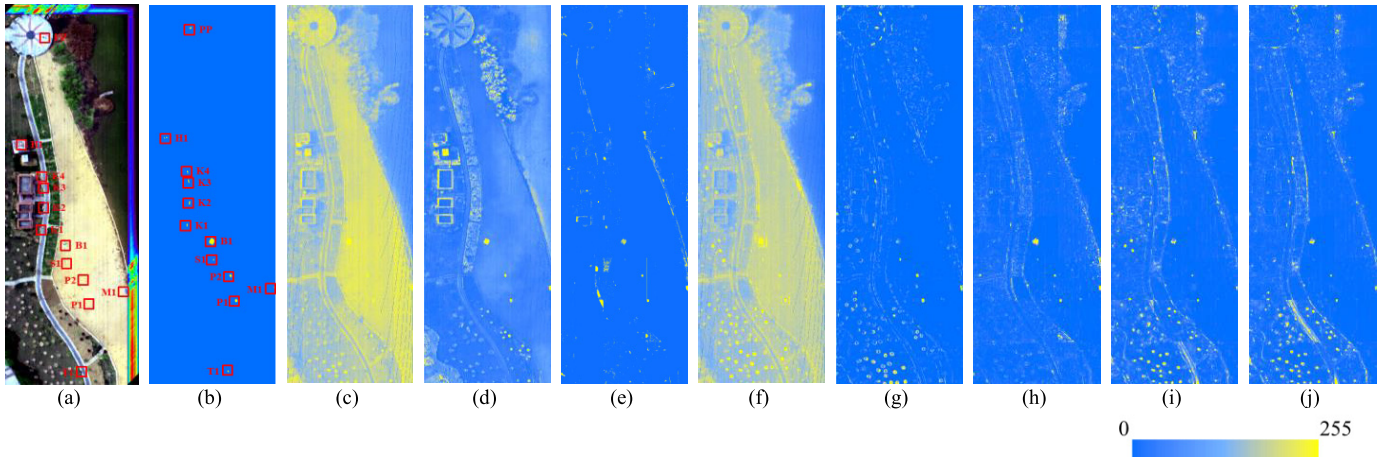


Fig. 6. UAV-borne WHU-Hi-Park dataset and the anomaly detection results. (a) Data cube. (b) Ground-truth map. (c) GRX. (d) LRASR. (e) ADLR. (f) CRD. (g) SC_AAE. (h) Auto-AD. (i) W-FCAE. (j) DeepLR.

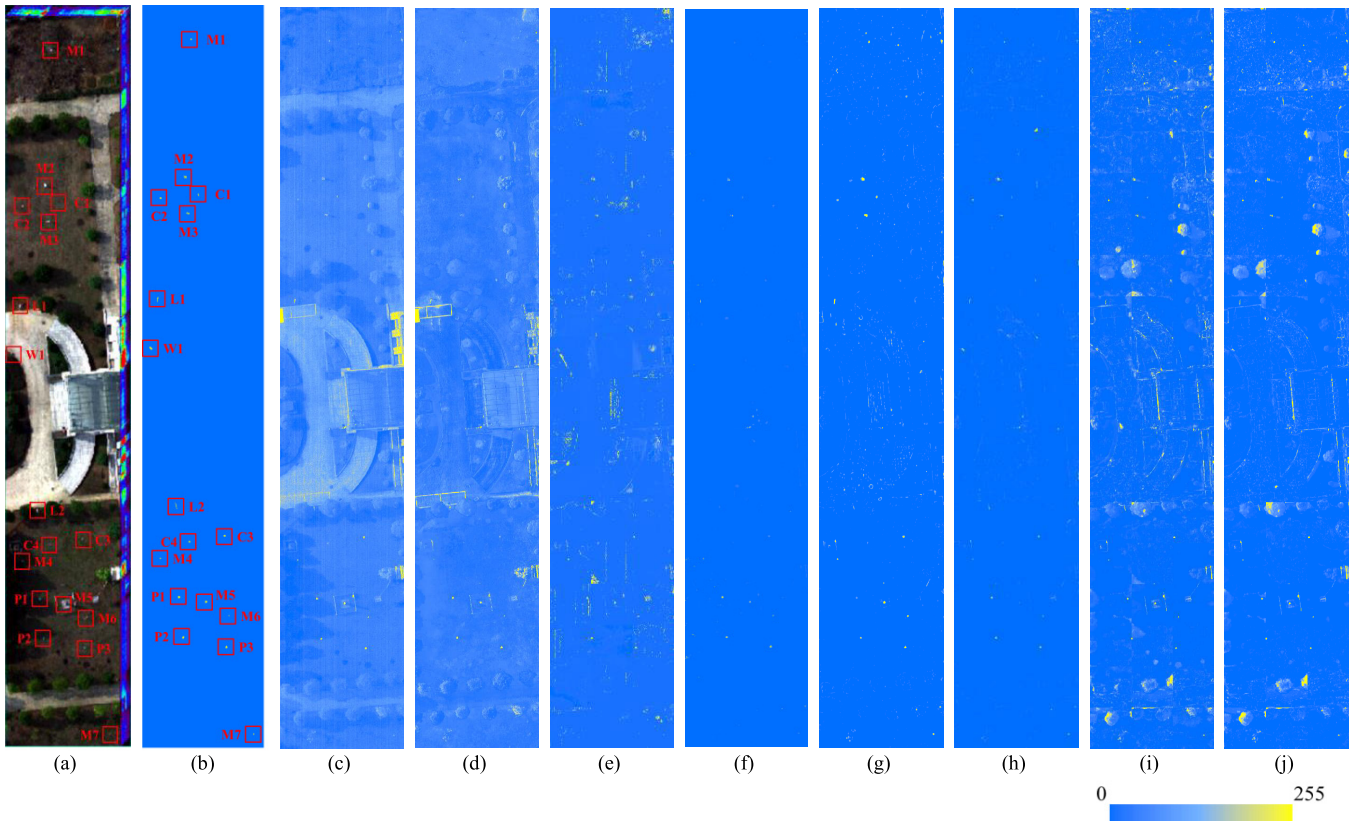


Fig. 7. UAV-borne WHU-Hi-Station dataset and the anomaly detection results. (a) Data cube. (b) Ground-truth map. (c) GRX. (d) LRASR. (e) ADLR. (f) CRD. (g) SC_AAE. (h) Auto-AD. (i) W-FCAE. (j) DeepLR.

reconstruction performance for the edges since there are fewer false alarms in the edges.

3) UAV-Borne WHU-Hi-Park Dataset: For the WHU-Hi-Park dataset, as shown in Fig. 6, the result of DeepLR shows fewer false alarms than the results of GRX, LRASR, and CRD. The background is obvious in the results of GRX, LRASR, and CRD, whereas it is hardly visible in the result of DeepLR. In the result of ADLR, the false alarms and noise exhibit very high responses. The visual effects of the results of SC_AAE and DeepLR are similar. The result of Auto-AD has less false alarms, and the anomalies also exhibit strong responses. Compared with the result of W-FCAE, there is less salt-and-pepper noise in the result of DeepLR, and the anomalies show stronger responses. In addition, at the base of the detection maps, the result of DeepLR shows fewer false alarms than the result of W-FCAE, which means that DeepLR achieves a more precise background reconstruction, thus demonstrating the effectiveness of the low-rank prior.

4) UAV-Borne WHU-Hi-Station Dataset: The detection results for the WHU-Hi-Station dataset are shown in Fig. 7.

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The background in the result of DeepLR shows a weaker response than that in the results of GRX and LRASR, and the anomalies show a higher contrast with the background. Actually, the false alarms and noise are with very high responses in the results of ADLR. Although few false alarms exist in the results of CRD and Auto-AD, the responses of the anomalies are not very strong. From the visual effect, the result of SC_AAE exhibits fewer false alarms. Compared with the results of W-FCAE and DeepLR, there is less salt-and-pepper noise in the result of DeepLR. With the introduction of the low-rank prior, the proposed DeepLR method exhibits superior performance in background reconstruction on all four datasets, which reflects the advantage of the framework combining the model- and data-driven approaches.

C. Quantitative Evaluation

The multiparameter receiver operating characteristic (3-D ROC) curves [61], [62] are provided for the quantitative evaluation. The 3-D ROC curves are plotted for three parameters: the probability of detection P_D, the false alarm rate P_F, and the threshold τ. In addition to the conventional 2-D ROC curve (P_D, P_F) [63], two new 2-D ROC curves can be derived from the 3-D ROC curve, i.e., the curves plotted by (P_D, τ) and (P_F, τ), respectively. Accordingly, the area under the curve (AUC) [64] is employed for the overall evaluation of the detection performance. In addition to the AUC scores of the ROC curve (P_D, P_F) ($\mathrm{AUC}_{(D,F)}$), the AUC score of the ROC curve (P_D, τ) ($\mathrm{AUC}_{(D,\tau)}$), and the AUC score of the ROC curve (P_F, τ) ($\mathrm{AUC}_{(F,\tau)}$), the other four measures are derived from 3-D ROC curves, including the target detectability of a detector ($\mathrm{AUC_{TD}}$), the background suppressibility of a detector ($\mathrm{AUC_{BS}}$), the overall detection probability ($\mathrm{AUC_{ODP}}$), and the signal-to-noise probability ratio ($\mathrm{AUC_{SNPR}}$) [62]. A larger AUC score for the curve (P_D, P_F) and (P_D, τ) indicates a better detection performance. A larger $\mathrm{AUC_{TD}}$ indicates a better target detectability of a detector. A smaller $\mathrm{AUC}_{(F,\tau)}$ or a larger $\mathrm{AUC_{BS}}$ demonstrates a better background suppression performance. A larger $\mathrm{AUC_{ODP}}$ indicates a higher overall detection probability of a detector. A larger $\mathrm{AUC_{SNPR}}$ indicates a higher signal-to-noise probability ratio.

To further assess the effect of the background suppression, separability maps are also provided. The detection maps are normalized in the range of [0, 255], and the distributions of the anomalous pixels' values and the background pixels' values can be compared via the box charts. As shown in Fig. 10, the green box denotes the anomalous pixels' values in the range of [10%, 90%], and the red box denotes the background. The lines at the top and bottom of each box represent the extreme values, and the median line denotes the mid value of the pixels' values. The background is effectively suppressed when the background box is narrow and the anomaly box is separated from the background box.

The 3-D ROC curves are shown in Fig. 8. The 2-D ROC curves are shown in Figs. 9–12. The AUC scores are listed in Tables V–VIII where the figures in bold are the best results and the underlined figures are the second best results. The separability maps are shown in Fig. 13.

TABLE IV
EXECUTION TIMES

	HYDICE	AVIRIS-1	WHU-Hi-Park	WHU-Hi-Station
GRX	0.14	0.33	51.91	71.96
LRASR	23.48	28.81	19962.79	52517.13
ADLR	49.32	30.46	25405.03	34227.51
CRD	60.31	79.47	37427.6	37181.45
SC_AAE	120.36	130.56	9944.3296	21728.03
Auto-AD	67.81	154.16	2604.76	4730.52
W-FCAE	45.20	25.11	617.96	898.19
DeepLR	297.68	159.59	10187.13	13714.35

1) Airborne HYDICE Dataset: For the airborne HYDICE dataset, the 3-D ROC curves are shown in Fig. 8(a), the 2-D ROC curves are shown in Fig. 9, and the AUC scores are listed in Table V. For the 2-D ROC curves (P_D, P_F), the curve of DeepLR is always above those of GRX, ADLR, and SC_AAE. Although the curves of LRASR, CRD, Auto-AD, and W-FCAE intersect with the curve of DeepLR, DeepLR exhibits a much lower false alarm rate when the probability of detection reaches beyond about 80%. When achieving a 100% probability of detection, the false alarm rates of GRX, LRASR, ADLR, CRD, SC_AAE, Auto-AD, W-FCAE, and DeepLR are 2%, 17.06%, 24.21%, 1%, 1.25%, 0.8%, 2.31%, and 0.4%, respectively. In addition, DeepLR obtains the highest $\mathrm{AUC}_{(D,F)}$ of 0.9996, which is very close to 1. The detection performance of DeepLR shows an improvement over that of W-FCAE. Such an improvement in detection performance comes from the low-rank prior since it is the only difference between these two methods. Although SC_AAE achieves the highest $\mathrm{AUC}_{(D,\tau)}$ and $\mathrm{AUC_{TD}}$, it has the highest $\mathrm{AUC}_{(F,\tau)}$ and the lowest $\mathrm{AUC_{BS}}$. It means that SC_AAE has better target detectability, while its background suppressibility is below the average. W-FCAE achieves the highest $\mathrm{AUC_{SNPR}}$. The $\mathrm{AUC_{ODP}}$ of each method is not very separated.

The separability map for the airborne HYDICE dataset is shown in Fig. 13(a). Except for ADLR, the anomaly box is separated from the background box for the other methods. The anomaly box of DeepLR shows greater degree of separation from the background box than W-FCAE. This indicates that DeepLR achieves a better background suppression performance than W-FCAE with the introduction low-rank prior.

The anomalies in the HYDICE dataset are with very small sizes, which cannot demonstrate the advantage of DeepLR in the detection of anomalies with larger sizes. Comparing DeepLR, Auto-AD, and W-FCAE, DeepLR exhibits good target detectability since its $\mathrm{AUC}_{(D,\tau)}$, $\mathrm{AUC_{TD}}$, and $\mathrm{AUC_{ODP}}$ are all the second highest, whereas the background suppressibility of Auto-AD and W-FCAE is better than DeepLR.

2) Airborne AVIRIS-1 Dataset: For the airborne AVIRIS-1 dataset, the 3-D ROC curves are shown in Fig. 8(b), the 2-D ROC curves are plotted in Fig. 10, and the AUC

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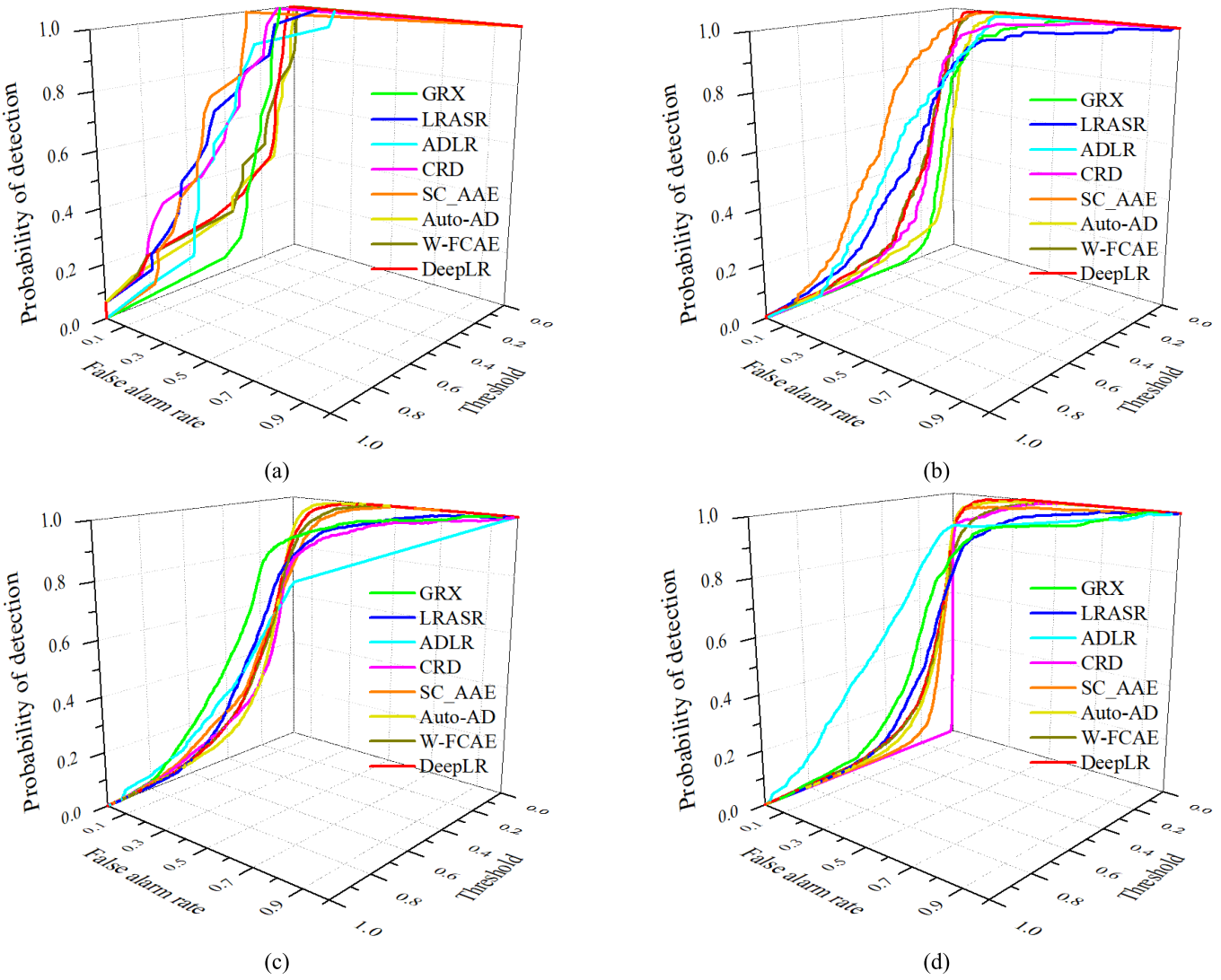


Fig. 8. 3-D ROC curves. (a) HYDICE dataset. (b) AVIRIS-1 dataset. (c) WHU-Hi-Park dataset. (d) WHU-Hi-Station dataset.

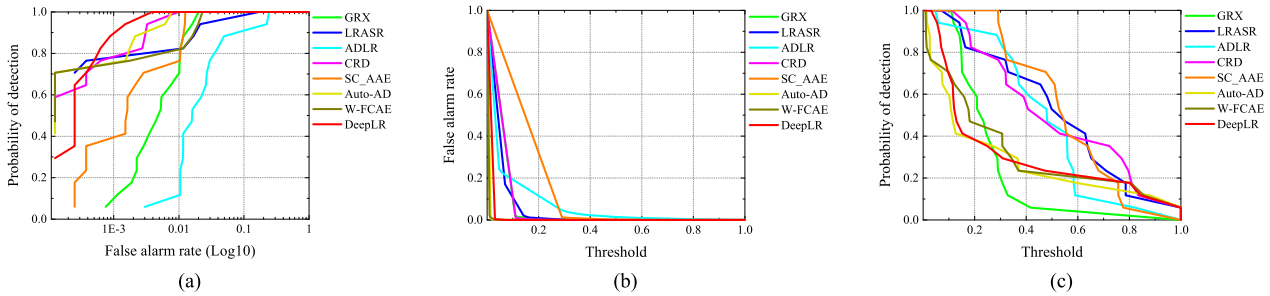


Fig. 9. 2-D ROC curves of HYDICE dataset. (a) ROC curve of (P_D, P_F). (b) ROC curve of (P_F, τ). (c) ROC curve of (P_D, τ).

scores are listed in Table VI. For the curves of (P_D, P_F), the curve of DeepLR is nearly always above those of the counterpart methods, except for LRASR. When achieving a 100% probability of detection, the false alarm rates of GRX, LRASR, ADLR, CRD, SC_AAE, Auto-AD, W-FCAE, and DeepLR are 54%, 90%, 38%, 62%, 15%, 22%, 12%, and 9%, respectively. DeepLR also obtains the highest $\mathrm{AUC}_{(D,F)}$ of 0.9845. Comparing DeepLR and W-FCAE, the background suppression of DeepLR is mainly reflected in some small false alarms and noise, which will not significantly improve the overall detection accuracy. On this dataset, deep learning-based SC_AAE achieves the highest $\mathrm{AUC}_{\mathrm{ODP}}$, $\mathrm{AUC}_{(D,\tau)}$, and $\mathrm{AUC}_{\mathrm{TD}}$. Auto-AD achieves the lowest $\mathrm{AUC}_{(F,\tau)}$. DeepLR obtains the highest $\mathrm{AUC}_{\mathrm{BS}}$ and the highest $\mathrm{AUC}_{\mathrm{SNPR}}$.

As shown in Fig. 13(b), the anomaly box is separated from the background box only for the three deep learning-based

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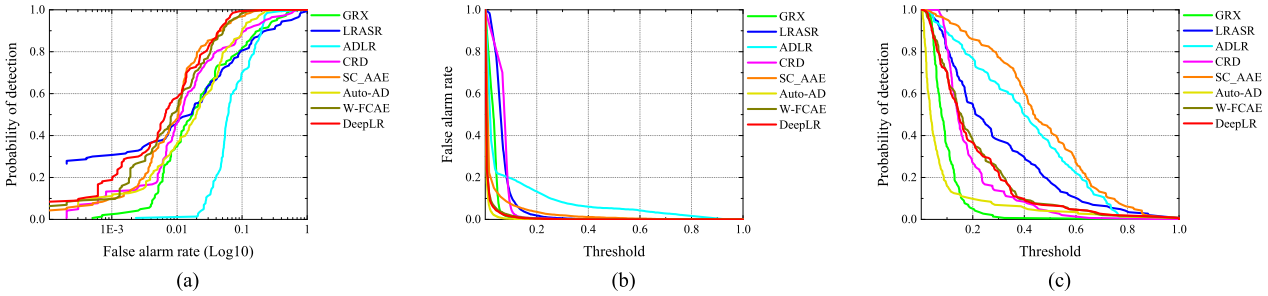


Fig. 10. 2-D ROC curves of AVIRIS-1 dataset. (a) ROC curve of (P_D, P_F). (b) ROC curve of (P_F, τ). (c) ROC curve of (P_D, τ).

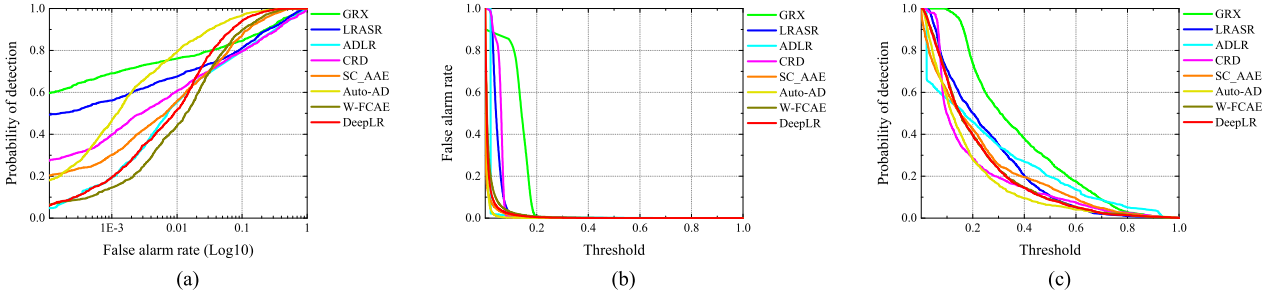


Fig. 11. 2-D ROC curves of WHU-Hi-Park dataset. (a) ROC curve of (P_D, P_F). (b) ROC curve of (P_F, τ). (c) ROC curve of (P_D, τ).

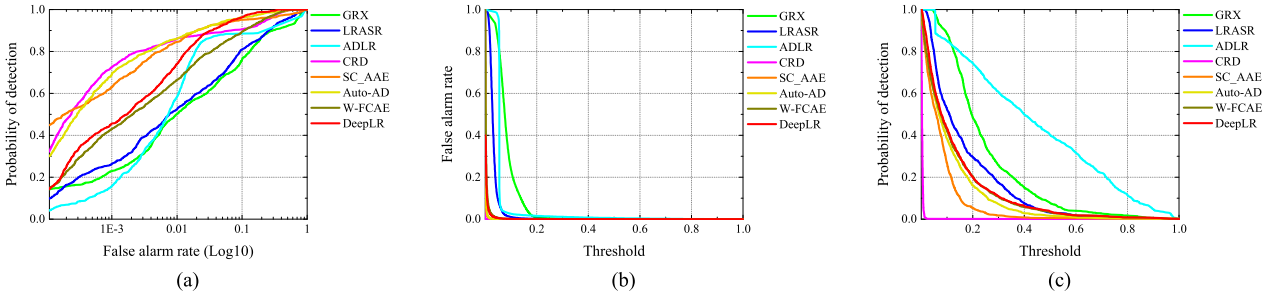


Fig. 12. 2-D ROC curves of WHU-Hi-Station dataset. (a) ROC curve of (*PD*, *PF*). (b) ROC curve of (*PF*, τ). (c) ROC curve of (*PD*, τ).

TABLE V
AUC SCORES FOR THE HYDICE DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,r)}$	$AUC_{(F,r)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}	AUC_{SNPR}
GRX	0.9938	0.2487	0.0571	1.2425	0.9367	1.1916	4.3555
LRASR	0.9920	0.5189	0.0490	1.5109	0.9430	**1.4699**	10.5898
ADLR	0.9624	0.4640	0.0713	1.4264	0.8911	1.3927	6.5077
CRD	0.9991	0.5145	0.0567	1.5136	0.9424	1.4578	9.0741
SC_AAE	0.9962	**0.5458**	0.1487	**1.5420**	0.8475	1.3971	3.6705
Auto-AD	0.9991	0.2756	0.0070	1.2747	**0.9921**	1.2686	39.3714
W-FCAE	0.9964	0.3115	**0.0051**	1.3079	0.9913	1.3064	**61.0784**
DeepLR	**0.9996**	0.3054	0.0156	1.3050	0.9840	1.2898	19.5769

methods, SC_AAE, W-FCAE, and DeepLR, where DeepLR exhibits the better background suppressibility since it has the highest AUC_{BS}. It also demonstrates that the background reconstruction is more accurate with the introduction of low-rank prior.

AVIRIS-1 dataset has the similar size with HYDICE dataset, but the proportion of anomalies is much larger. It can be seen that the overall detection performance index $AUC_{(D,F)}$ is obviously decreased for Auto-AD and W-FCAE, which are based only on the adaptive-weighting strategy. DeepLR introduces low-rank prior to ensure the detection ability of anomalies with larger sizes. Comparing DeepLR, Auto-AD, and W-RCAE, DeepLR achieves the highest or the second highest scores in all indicators and exhibits a balance between the target detectability and the background suppressibility.

3) UAV-Borne WHU-Hi-Park Dataset: For the UAV-borne WHU-Hi-Park dataset, the 3-D ROC curves are shown in Fig. 8(c), the 2-D ROC curves are plotted in Fig. 11, and

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TABLE VI
AUC SCORES FOR THE AVIRIS-1 DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}	AUC_{SNPR}
GRX	0.9370	0.0968	0.0309	1.0338	0.9061	1.0659	3.1327
LRASR	0.9146	0.2956	0.0665	1.2102	0.8481	1.2291	4.4451
ADLR	0.9081	0.3997	0.0852	1.3078	0.8229	1.3145	4.6913
CRD	0.9530	0.1857	0.0686	1.1387	0.8844	1.1171	2.7070
SC_AAE	0.9820	**0.4607**	0.0307	**1.4427**	0.9513	**1.4300**	15.0065
Auto-AD	0.9628	0.0884	**0.0053**	1.0512	0.9575	1.0831	16.6792
W-FCAE	0.9812	0.2058	0.0116	1.1870	0.9696	1.1942	17.7414
DeepLR	**0.9845**	0.2013	0.0098	1.1858	**0.9747**	1.1915	**20.5408**

TABLE VII
AUC SCORES FOR THE WHU-HI-PARK DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}	AUC_{SNPR}
GRX	0.9379	**0.3712**	0.1280	**1.3091**	0.8099	**1.2432**	2.9000
LRASR	0.9274	0.2469	0.0463	1.1743	0.8811	1.2006	5.3326
ADLR	0.8234	0.2550	0.0239	1.0784	0.7995	1.2311	10.6695
CRD	0.9095	0.1951	0.0581	1.1046	0.8514	1.1370	3.3580
SC_AAE	0.9579	0.2219	0.0070	1.1798	0.9509	1.2149	**31.7000**
Auto-AD	**0.9864**	0.1728	**0.0067**	1.1592	**0.9797**	1.1661	25.7910
W-FCAE	0.9610	0.2118	0.0167	1.1728	0.9443	1.1951	12.6826
DeepLR	0.9736	0.2101	0.0129	1.1837	0.9607	1.1972	16.2868

TABLE VIII
AUC SCORES FOR THE WHU-HI-STATION DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}	AUC_{SNPR}
GRX	0.8988	0.2453	0.0825	1.1441	0.8163	1.1628	2.9733
LRASR	0.9202	0.1627	0.0316	1.0829	0.8886	1.1311	5.1487
ADLR	0.9260	**0.4306**	0.0610	**1.3566**	0.8650	**1.3696**	7.0590
CRD	0.9722	0.0041	**0.0003**	0.9763	0.9719	1.0038	13.6667
SC_AAE	0.9708	0.0747	0.0007	1.0455	0.9701	1.0740	**106.7143**
Auto-AD	0.9852	0.1094	0.0015	1.0946	**0.9837**	1.1079	72.9333
W-FCAE	0.9660	0.1269	0.0039	1.0929	0.9621	1.1230	32.5385
DeepLR	**0.9853**	0.1316	0.0028	1.1169	0.9825	1.1288	47.0000

the AUC scores are listed in Table VII. This dataset (600 × 1800 pixels) is much larger than the two airborne datasets. The enhanced image size and complexity of the background greatly increase the difficulty of anomaly detection. The curve (P_D, P_F) of DeepLR is nearly always above those of ADLR and W-FCAE. Although the curves of GRX, LRASR, CRD, SC_AAE, and DeepLR are intersected, DeepLR exhibits a much lower false alarm rate when the detection of probability increases beyond about 80%. When achieving a 100% probability of detection, the false alarm rates of GRX, LRASR, ADLR, CRD, SC_AAE, Auto-AD, W-FCAE, and DeepLR are 86%, 98%, 100%, 99%, 77%, 58%, 55%, and 48%, respectively. The curve of (P_D, P_F) Auto-AD is above that of DeepLR, and Auto-AD achieves the highest $AUC_{(D,F)}$ of 0.9864. Compared with the results obtained for the airborne HYDICE dataset, the detection accuracy of GRX, RASR, ADLR, and CRD shows an obvious decrease, which can be attributed to either the overly idealistic statistical assumption or the poorly generalized parameter settings.

GRX achieves higher target detectability since it has the highest AUC_{ODP}, $AUC_{(D,\tau)}$, and AUC_{TD}, while its background suppressibility is not good since it has the highest $AUC_{(F,\tau)}$ and second-lowest AUC_{BS}. SC_AAE achieves the highest AUC_{SNPR}. Auto-AD has the lowest $AUC_{(F,\tau)}$ and the highest AUC_{BS}, which means that Auto-AD achieves the best background suppression performance. As shown in Fig. 13(c), the anomaly boxes are not separated from the background boxes for SC_AAE, whereas the anomaly box and background box are separated for DeepLR, which means that DeepLR has a better background suppressibility than SC_AAE.

WHU-Hi-Park dataset is much larger than HYDICE and AVIRS-1 datasets and contains anomalies with different sizes. DeepLR incorporates low-rank prior and adaptive-weighting strategy to ensure the detection ability of anomalies with different sizes. Comparing DeepLR, Auto-AD, and W-FCAE, DeepLR achieves the balance between the target detectability and the background suppressibility, and all the AUC scores reach to the highest or the second highest level.

4) UAV-Borne WHU-Hi-Station Dataset: For the UAV-borne WHU-Hi-Station dataset, the 3-D ROC curves are shown in Fig. 8(d), the 2-D ROC curves are plotted in Fig. 12, and the AUC scores are listed in Table VIII. This dataset (3600 × 600 pixels) is larger than the UAV-borne WHU-Hi-Park dataset (600 × 1800 pixels). The ROC curve (P_D, P_F) of DeepLR is nearly always above those of GRX, LRASR, ADLR, and W-FCAE, whereas the curve of DeepLR is intersected with those of CRD, SC_AAE, and Auto-AD. When achieving a 100% probability of detection, the false alarm rates of GRX, LRASR, ADLR, CRD, SC_AAE, Auto-AD, W-FCAE, and DeepLR are 91%, 99%, 98%, 59%, 92%, 65%, 70%, and 34%, respectively. DeepLR also obtains the highest $AUC_{(D,F)}$ on this dataset. The improved detection

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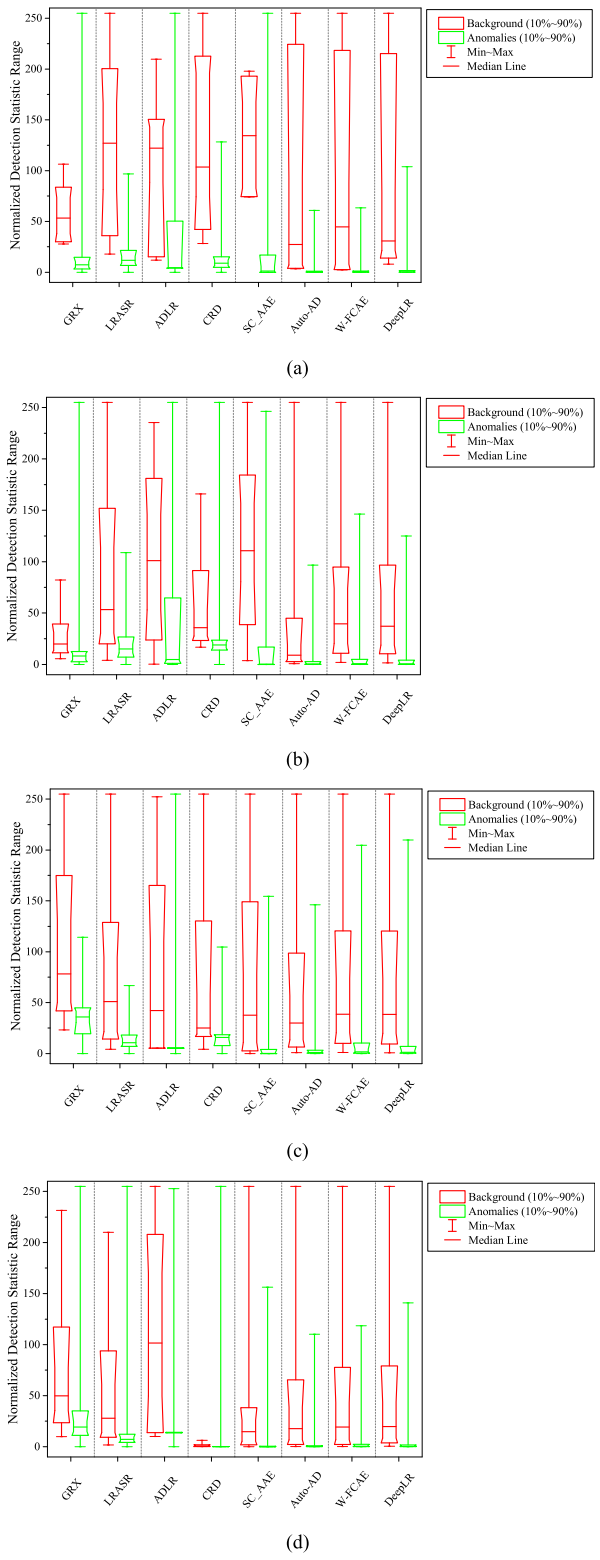


Fig. 13. Separability maps. (a) Airborne HYDICE dataset. (b) Airborne AVIRIS-1 dataset. (c) UAV-borne WHU-Hi-Park dataset. (d) UAV-borne WHU-Hi-Station dataset.

performance of DeepLR over W-FCAE on the UAV-borne WHU-Hi-Station dataset further demonstrates the effectiveness of the low-rank prior.

Although ADLR achieves the highest $\mathrm{AUC}_{\mathrm{ODP}}$, $\mathrm{AUC}_{(D,\tau)}$, and $\mathrm{AUC}_{\mathrm{TD}}$, it cannot balance the target detectability and the background suppressibility. SC_AAE achieves the highest $\mathrm{AUC}_{\mathrm{SNPR}}$. Auto-AD achieves the highest $\mathrm{AUC}_{\mathrm{BS}}$. The proposed DeepLR method exhibits a balance between the target detectability and the background suppressibility on the experimental four datasets. The separability map is shown in Fig. 13(d). For the GRX, LRASR, and ADLR methods, the anomaly boxes and background boxes are not separated. The CRD, SC_AAE, Auto-AD, W-FCAE, and DeepLR methods exhibit better background suppression effects since the background boxes are suppressed over a small range, among which DeepLR obtains the second-highest $\mathrm{AUC}_{\mathrm{BS}}$.

The execution times of the different methods are listed in Table IV. The GRX method is the most time efficient since the model is the simplest. With the increase of the data volume, the execution times of the LRASR, ADLR, and CRD methods are significantly increased. The execution time of the proposed DeepLR method is greater than that of the W-FCAE method due to the additional procedure of low-rank background estimation. However, compared with the LRASR, ADLR, and CRD methods based on image priors, the increase is much smaller when the data volume is increased.

On the WHU-Hi-Station dataset with larger size, DeepLR also achieves the balance between the target detectability and the background suppressibility, compared with Auto-AD and W-FCAE. DeepLR has larger $\mathrm{AUC}_{(D,\tau)}$ and $\mathrm{AUC}_{\mathrm{TD}}$ than Auto-AD and W-FCAE and achieves the second best $\mathrm{AUC}_{(F,\tau)}$ and $\mathrm{AUC}_{\mathrm{BS}}$ among the results of DeepLR, Auto-AD, and W-FCAE. These factors demonstrate the effectiveness of the low-rank prior.

5) Additional Public Datasets: To further test the performance of the proposed DeepLR method, experiments on several other public datasets were carried out. The public datasets are listed in Fig. 14. The AVIRIS-2 dataset [65] is a subset of an image collected by the AVIRIS sensor. The second dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) [66] over Pavia city center in northern Italy, with 102 spectral bands covering the spectral range of 430–860 nm. The image is of 108 × 120 pixels, and four vehicles on the bridge and the bare soil near the bridge pier are considered as anomalies, with a total number of 43 pixels, accounting for 0.33% of the image. The Cri dataset [18] covers the spectral range of 650–1000 nm, with 46 spectral bands and 400 × 400 pixels, in which the rocks are the anomalies, which account for 1.38% of the image.

The AUC scores of the different methods are listed in Table IX, where the proposed DeepLR method achieves the highest AUC scores for the ROSIS and Cri datasets. Among the different methods, only the DeepLR method achieves an AUC score of greater than 0.9 on all four datasets. Furthermore, the proposed DeepLR method shows superior robustness on these four public datasets, as shown in Fig. 15, where DeepLR exhibits small variation of detection accuracy on all the experimental datasets.

D. Discussion

1) Effectiveness of the Adaptive-Weighting Strategy: To evaluate the effectiveness of the adaptive-weighting strategy,

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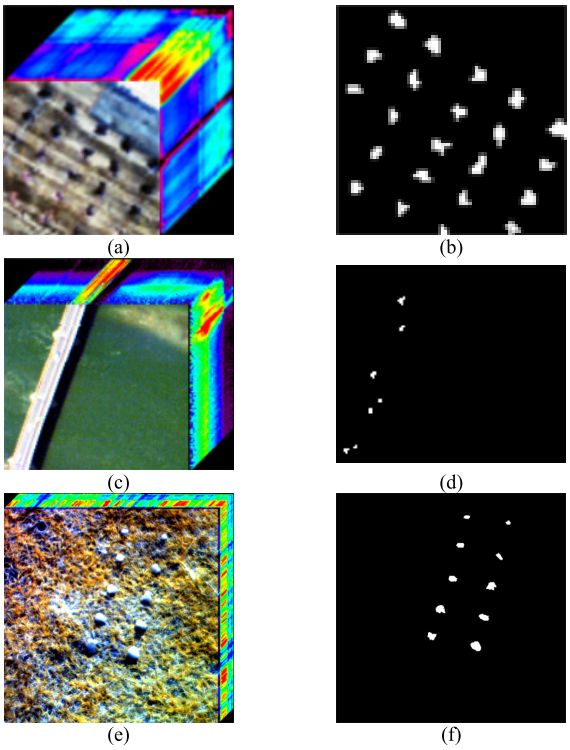


Fig. 14. Additional public datasets. AVIRIS-2 dataset: (a) data cube and (b) ground-truth map. ROSIS dataset: (c) data cube and (d) ground-truth map. Cri dataset: (e) data cube and (f) ground-truth map.

TABLE IX
AUC SCORES FOR THE ADDITIONAL PUBLIC DATASETS

	AVIRIS-2	ROSIS	Cri
GRX	0.7521	0.9983	0.9678
LRASR	0.8097	0.9801	0.8652
ADLR	0.8918	0.9981	0.9579
CRD	0.8729	0.9842	0.9186
SC_AAE	**0.9496**	0.9984	0.8849
Auto-AD	0.8939	0.9983	0.9643
W-FCAE	0.8962	0.9996	0.9765
DeepLR	0.9163	**0.9997**	**0.9815**

TABLE X
COMPARISON EXPERIMENTS FOR THE ADAPTIVE-WEIGHTING STRATEGY

Method	AUC score
Without the low-rank prior and weighting	0.9939
Without the low-rank prior but with the weighting	0.9964
With the low-rank prior but without the weighting	0.9977
With the low-rank prior and weighting	**0.9996**

comparison experiments were conducted on the HYDICE dataset, in which the anomalies are very small. The experimental settings and AUC scores are shown in Table X, where the detection accuracy is improved with the introduction of the adaptive-weighting strategy, both with and without the constraint of the low-rank prior. The experimental results confirm the effectiveness of the adaptive-weighting strategy.

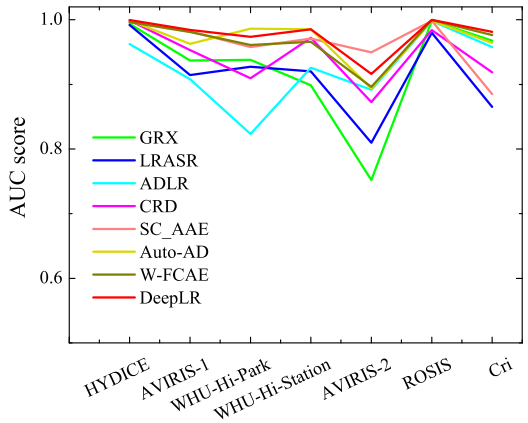


Fig. 15. Line chart of the AUC scores.

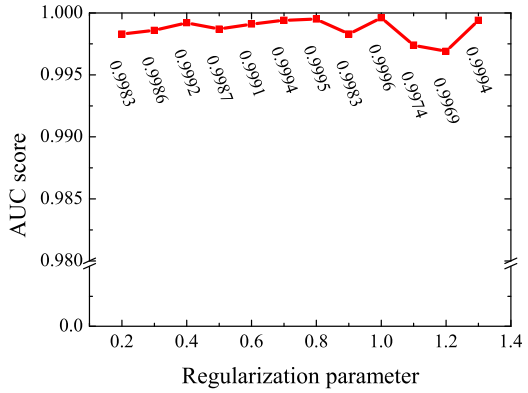


Fig. 16. Sensitivity analysis for parameter λ on the HYDICE dataset.

2) Sensitivity Analysis for the Regularization Parameter λ: A sensitivity analysis was conducted on the HYDICE dataset to investigate the impact of regularization parameter λ on the detection performance. As shown in Fig. 16, the AUC score exhibits the trend of first improving and then decreasing as regularization parameter λ is increased. The best result is obtained when parameter λ equals 1.0. Therefore, the experiments in this study were conducted with the regularization parameter λ set as 1.0 for all the experimental datasets.

V. CONCLUSION

In this article, a DeepLR has been proposed for hyperspectral anomaly detection by combining the respective advantages of a model-driven low-rank prior and a data-driven AE. To be specific, the low-rank prior of the background is introduced into a fully convolutional AE with skip connections, where the background can be approximately reconstructed, while the anomalies are reconstructed with significant reconstruction errors; thus, the reconstruction errors indicate the anomalous degree.

The proposed DeepLR method avoids the preprocessing and detector construction procedures and realizes end-to-end hyperspectral anomaly detection, from the input hyperspectral image to the final anomaly detection result. Experiments on several public datasets and two large UAV-borne hyperspectral datasets confirmed the effectiveness of the introduction of the

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low-rank prior and showed that the proposed DeepLR method has a better generalization capability than the comparative methods.

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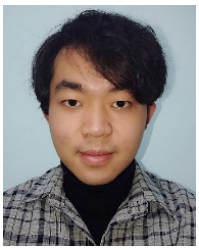
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